MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 59-CONDUCTOR, TYPE MSPW

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (watertight)

Conductor size 22-7 (16 required):

First - Copper conductor, in accordance with ASTM B 286, size 22-7, tin coated.
Second - Fluorocarbon resin insulation, nominal thickness 0.044 inch. (Taped insulation may be used.)
Third - Braided shield of AWG no. 34 or no. 36 copper. Minimum coverage 85 percent; braid angle 30 to 35 degrees.
Fourth - Shield insulation of one polyester tape, type G in accordance with MIL-I-631, opaque white, with a jacket of transparent polyamide in accordance with ASTM D 4066. Nominal thickness 0.003.
- Alternate shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Inner tape opaque white; outer tape yellow.

For either type of shield insulation, use standard identification code applied by method 2, on the inner tape, using numbers 30 through 45.

Conductor size 20-7 (18 required):

First - Copper conductor, in accordance with ASTM B 286 size 20-7, tin coated.
Second - Polyvinyl chloride insulation (105°C), colored white. Nominal thickness 0.022 inch.
Third - Insulation jacket of clear polyamide in accordance with ASTM D 4066. Nominal thickness 0.003 inch.

FSC 6145

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Fourth - Braided shield of AWG no. 34 or 36 copper, tin coated. Minimum coverage of 85 percent; braid angle of 30 to 35 degrees.

Fifth - Shield insulation consisting of one polyester tape, type G in accordance with MIL-I-631, with a jacket of clear polyamide. Nominal thickness 0.003 inch.
- Alternate shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed.

For either type of shield insulation, use standard identification code applied by method 2, on the inner tape, using numbers 12 through 29.

Sixth - Conductors numbered 12, 13 and 14 formed into a triad. Nominal lay 3 inches.

Seventh - Conductors numbered 19 through 24 formed into three pairs, with the following pairing: numbers 19 and 20; numbers 21 and 22; numbers 23 and 24. Nominal lay 3 inches.

Conductor size AWG 16 (25 required):

First - Copper conductor, AWG size 16, class B stranding, tin coated.

Second - Polyvinyl chloride insulation (105°C). Nominal wall thickness 0.014 inch. Eleven conductors colored white; eleven conductors colored black; three conductors colored red.

Third - Jacket of clear polyamide in accordance with ASTM D 4066, over each conductor. Nominal thickness 0.003 inch.

Fourth - Sixteen conductors formed into eight pairs, each pair consisting of one white and one black conductor. Nominal lay 3 inches.

Fifth - Nine conductors formed into three triads, each triad consisting of one white, one red and one black conductor. Nominal lay 3-1/2 inches.

Sixth - Over each pair and each triad, a braided shield of AWG no. 34 or no. 36 copper, tin coated. Minimum coverage 85 percent; braid angle of 30 to 35 degrees.

Seventh - Shield insulation, over each shield, consisting of one polyester tape, type G in accordance with MIL-I-631, with a jacket of clear polyamide. Nominal thickness 0.003 inch.
- Alternate shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed.

For either type of insulation, use standard identification code applied by method 2, on the inner tape. The triads shall be numbered 1, 2 and 3; the pairs shall be numbered 4 through 11.

Assembly:

First - A solid polychloroprene filler as the center of the core. Nonfibrous fillers shall be employed throughout the cable.

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Second - Inner layer, consisting of the following:
(1) The shielded triads numbered 1, 2 and 3.
(2) The triad consisting of the conductors numbered 12, 13 and 14.
(3) The conductors numbered 15, 16, 17 and 18.
Third - A binder tape.
Fourth - Middle layer, consisting of the following:
(1) The eight shielded pairs numbered 4 through 11.
(2) The three pairs consisting of conductors numbered 19 and 20, 21 and 22, 23 and 24.
Fifth - A binder tape.
Sixth - Outer layer, consisting of the following:
(1) The conductors numbered 25 through 45, cabled consecutively.
Seventh - A binder tape.
Eighth - Polychloroprene jacket (black) nominal thickness 0.110 inch. Cable overall diameter 1.565-inches minimum, 1.635-inches maximum. Cable surface marking required.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	
Size 22 conductors	17.71
Size 20 conductors	11.03
Size AWG 16 conductors	4.44
Voltage withstand - volts, root mean square, minimum	
Size 22 conductors	
Conductor to conductor	5000
Conductor to shield	3000
Shield to shield	200
Size 20 and AWG 16 conductors	
Conductor to conductor	2000
Conductor to shield	1000
Shield to shield	200
Insulation resistance - megohms/1000 feet, minimum	
Size 22 conductors	
Conductor to conductor	5000
Conductor to shield	5000
Shield to shield	100
Size 20 and AWG 16 conductors	
Conductor to conductor	150
Conductor to shield	150
Shield to shield	100
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-915, table XXII, for limits of water leakage	No failure
Characteristic impedance (size 22 conductors only) - at 1 MHz; ohms	55 ± 5
Capacitance (size 22 conductors only) - at 1 kHz; pF/ft, maximum	27

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Drip - at 75 ± 1°C	Zero
Cold bending, cable - at 0 ± 2°C; over 18 inch maximum diameter mandrel	No damage
Physicals (unaged)	
Insulation (polyvinyl chloride only (PVC))	
Tensile strength - lbf/in^2, minimum	1800
Elongation - percent, minimum	100
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1800
Elongation - percent, minimum	300
Set - inch, maximum	30/64
Group C:	
Flammability - inches, maximum	2
Physicals (aged)	
Insulation (polyvinyl chloride only (PVC))	
Air oven	
Elongation - percent of unaged, minimum	90
Cable jacket	
Air pressure heat	
Tensile strength - percent of unaged, minimum	65
Elongation - percent of unaged, minimum	65
Hot oil immersion	
Tensile strength - percent of unaged, minimum	65
Elongation - percent of unaged, minimum	65
Permanence of printing (jacket) - cycles, minimum	250
Cable filler removability	No failure
Shield-conformance to material coverage and construction	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Cable aging and compatibility (95 ± 3°C)	No failure

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UNIT ORDERING LENGTH: 500 feet (nominal).

Revision letters are not used to denote changes due to the extensiveness of the changes.

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

Preparing activity:
Navy - SH
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